Supplement S1: Mathematical Foundations

E_8 Exceptional Lie Algebra, G_2 Holonomy Manifolds, and K_7 Construction Brieuc de La Fournière

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Mathematical background for the K_7 framework: the E_8 and G_2 data the dictionary uses, the topological status of K_7, and the algebraic reference form.

Lean: the arithmetic statements of this supplement are formalized in Arithmon/K7-Lean (the library rests on fourteen axioms beyond Mathlib's foundations, four literature imports and ten interval certificates, and uses no `sorry`; the 33 relations themselves depend only on the standard axioms; main §8, S3.7). Formalization establishes the arithmetic. Lean names are given as they appear in the repository at commit `34ce857` (S3.7).

Abstract

This supplement presents the mathematical architecture underlying the K_7 framework. Part I gives the E_8 data the dictionary uses, with arithmetic rewritings of these data in the structural constants and their status. Part II introduces G_2 holonomy manifolds, including the characterization of the g_2 subalgebra as the kernel of the Lie derivative map. Part III discusses the topology of K_7: the pair $(b_2, b_3) = (21, 77)$ has no known construction. It does not appear among the catalogued compact G_2 manifolds, and a Joyce–Karigiannis ${\mathbb{Z}_2}^3$ route does not realise it, because the quotient cohomology it needs is incompatible with the group action (§8.4). Part IV describes the algebraic reference form, whose scale is fixed by the imposed normalization $\det(g) = 65/32$ (item R2 of the residue register, main §3.3, which lists what the framework sets rather than derives), and the status of Joyce's perturbation theorem for this setting: its smallness constant depends on geometric bounds that are not established here, and the existence of a compact torsion-free metric is open (main §9). The arithmetic statements are verified in Lean 4. Section 13 lists what changed since version 3.5.

Part 0: The Octonionic Foundation

0. Why This Framework Exists

The K_7 framework is motivated by a single algebraic fact:

The octonions $\mathbb{O}$ are the largest normed division algebra.

The chain of motivations, and where the framework's choices enter, is:

```
O (octonions, dim 8)
    |
    ▼
Im(O) = ℝ⁷ (imaginary octonions)
    |
    ▼
G₂ = Aut(O) (automorphism group, dim 14)
    |
    ▼
K₇ with G₂ holonomy (posited; a local metric is certified in [A]; global existence open)
    |
    ▼
Topological invariants (b₂, b₃) = (21, 77)   ← input (register item R1)
    |
    ▼
33 relations assigned to observables   ← dictionary (main §3)
```

Status: the relations are assignments, written in the constants that the steps above supply, and no geometric map from the invariants to the observables is known (main §3.2). The dataset (95 entries, of which the 33 relations are counted; they are its Type I) is described in S3, and the arithmetic of each relation in S2.

0.1 The Division Algebra Chain

Algebra	Dim	Physics Role	Stops?
$\mathbb{R}$	1	Classical mechanics	No
$\mathbb{C}$	2	Quantum mechanics	No
$\mathbb{H}$	4	Spin, Lorentz group	No
$\mathbb{O}$	**8**	**Exceptional structures**	**Yes**

The pattern terminates at $\mathbb{O}$. There is no 16-dimensional normed division algebra. The octonions are *the end of the line.*

0.2 G_2 as Octonionic Automorphisms

Definition: $G_2 = \{g \in GL(\mathbb{O}) : g(xy) = g(x)g(y) \text{ for all } x,y \in \mathbb{O}\}$

Property	Value	Role
$\dim(G_2)$	$14 = C(7,2) - C(7,1) = 21 - 7$	Q_{Koide} numerator
Action	Transitive on $S^6 \subset \text{Im}(\mathbb{O})$	Connects all directions
Embedding	$G_2 \subset SO(7)$	Preserves φ_0

0.3 Why $\dim(K_7) = 7$

The dimension 7 is a consequence of the octonionic structure:

- $\text{Im}(\mathbb{O})$ has dimension 7
- G_2 acts naturally on $\mathbb{R}^7$
- G_2 occurs as a holonomy group only in dimension 7 (Berger, §6.3)

The choice made by the framework is G_2 holonomy; dimension 7 then follows.

0.4 The Fano Plane: Combinatorial Structure of $\text{Im}(\mathbb{O})$

The 7 imaginary octonion units form the **Fano plane** PG(2,2), the smallest projective plane:

- 7 points (imaginary units $e_1 \dots e_7$)
- 7 lines (multiplication triples $e_i \times e_j = \pm e_k$)
- 3 points per line

Combinatorial counts:

- Point-line incidences: $7 \times 3 = 21 = C(7,2) = b_2$
- Automorphism group: PSL(2,7) with $|\text{PSL}(2,7)| = 168$

Arithmetic identity:

$$(b_3 + \dim(G_2)) + b_3 = 91 + 77 = 168 = |\text{PSL}(2,7)| = \text{rank}(E_8) \times b_2$$

It is an identity among the chosen integers of the kind recorded in S2.7; no geometric structure is attached to it.

Part I: E_8 Exceptional Lie Algebra

1. Root System and Dynkin Diagram

1.1 Basic Data

Property	Value	Role
Dimension	$\dim(\mathrm{E}_8) = 248$	Gauge DOF
Rank	$\mathrm{rank}(\mathrm{E}_8) = 8$	Cartan subalgebra
Number of roots	$\lvert\Phi(\mathrm{E}_8)\rvert = 240$	E_8 kissing number
Root length	$\sqrt{2}$	α_s numerator
Coxeter number	$h = 30$	Icosahedron edges
Dual Coxeter number	$h^\vee = 30$	McKay correspondence

1.2 Root System Construction

E_8 root system in $\mathbb{R}^8$ has 240 roots:

Type I (112 roots): Permutations and sign changes of $(\pm 1, \pm 1, 0, 0, 0, 0, 0, 0)$

Type II (128 roots): Half-integer coordinates with even minus signs:

$$\frac{1}{2}(\pm 1, \pm 1, \pm 1, \pm 1, \pm 1, \pm 1, \pm 1, \pm 1)$$

Verification: $112 + 128 = 240$ roots, all length $\sqrt{2}$.

Lean: the root-system statements are proved:

- `D8_roots_card` = 112, `HalfInt_roots_card` = 128
- `E8_roots_card` = 240, `E8_roots_decomposition`
- `E8_inner_integral`, `E8_norm_sq_even`, `E8_sub_closed`
- `E8_basis_generates`: Every lattice vector is integer combination of simple roots (theorem)

1.3 Cartan Matrix

$$A_{E_8} = \begin{pmatrix} 2 & 0 & -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 2 & 0 & -1 & 0 & 0 & 0 & 0 \\ -1 & 0 & 2 & -1 & 0 & 0 & 0 & 0 \\ 0 & -1 & -1 & 2 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & 2 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 & 2 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 & 2 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 & -1 & 2 \end{pmatrix}$$

Properties: $\det(A) = 1$ (unimodular), positive definite.

2. Weyl Group

2.1 Order and Factorization

$$|W(E_8)| = 696,729,600 = 2^{14} \times 3^5 \times 5^2 \times 7$$

2.2 Factorization in the Structural Constants

The Weyl group order can be written in the framework's structural constants:

$$|W(E_8)| = p_2^{\dim(G_2)} \times N_{gen}^{Weyl} \times Weyl^{p_2} \times \dim(K_7)$$

Factor	Exponent	Value	Origin
2^{14}	$\dim(G_2) = 14$	16384	$p_2^{\text{holonomy dim}}$
3^5	Weyl = 5	243	$N_{gen}^{\text{Weyl factor}}$
5^2	$p_2 = 2$	25	$\text{Weyl}^{\text{binary}}$
7^1	1	7	$\dim(K_7)$

This is an arithmetic rewriting of the prime factorization $2^{14} \cdot 3^5 \cdot 5^2 \cdot 7$, with each prime and exponent matched to a constant; it is the source of the constant Weyl = 5. Lean: `E8_weyl_order_factored`, `E8_weyl_factored` (the prime factorization).

2.3 Three Expressions for Weyl = 5

$$\frac{\dim(G_2) + 1}{N_{gen}} = \frac{b_2}{N_{gen}} - p_2 = \dim(G_2) - \text{rank}(E_8) - 1 = 5$$

Expression	Computation	Result
$(\dim(G_2) + 1)/N_{gen}$	$(14 + 1)/3$	5
$b_2/N_{gen} - p_2$	$21/3 - 2$	5
$\dim(G_2) - \text{rank}(E_8) - 1$	$14 - 8 - 1$	5

Three expressions in small integers that evaluate to the same small integer are an arithmetic fact about the chosen integers; they fix no further quantity (S2.1). Weyl enters $\det(g) = 65/32$ (R2) and the factorization of §2.2.

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3. Exceptional Chain

3.1 The Pattern

A pattern connects exceptional algebra dimensions to primes:

Algebra	n	dim(E_n)	Prime	Index
E_6	6	78	13	prime(6)
E_7	7	133	19	prime(8) = prime(rank(E_8))
E_8	8	248	31	prime(11) = prime(D_{bulk})

3.2 Three Factorizations

For n ∈ {6, 7, 8}:

$$\dim(E_n) = n \times prime(g(n))$$

where g(6) = 6, g(7) = rank(E_8) = 8, g(8) = D_{bulk} = 11 (= dim(M_4) + dim(K_7) = 4 + 7, the total bulk dimension of M-theory).

Check (Lean):

- E_6: 6 × 13 = 78 ✓
- E_7: 7 × 19 = 133 ✓
- E_8: 8 × 31 = 248 ✓

The index function g is chosen case by case (6, rank E_8, D_{bulk}), so the statement describes three factorizations and is not a structural theorem. Lean: `all_exceptional_chain_relations_certified`.

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4. $E_8 \times E_8$ Product Structure

4.1 Direct Sum

Property	Value
Dimension	496 = 248 × 2
Rank	16 = 8 × 2
Roots	480 = 240 × 2

4.2 τ Numerator Connection

The hierarchy parameter numerator:

$$\tau_{num} = 3472 = 7 \times 496 = \dim(K_7) \times \dim(E_8 \times E_8)$$

Lean: `tau_num_factorization`. τ is used only by the dimensionful entries of the dataset (Type II, S3).

4.3 The Binary Ratio $p_2 = 2$

p_2 is defined as $\dim(G_2)/\dim(K_7) = 14/7 = 2$ and is called the binary ratio. It is not a Pontryagin class [ERR, §6]. The number 2 also occurs as 496/248 and in the $\sqrt{2}$ of the E_8 root length; these are different occurrences of the number 2, not derivations of one constant.

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5. Exceptional Algebras from Octonions

The foundational role of octonions is established in Part 0. This section details the exceptional algebraic structures that emerge from $\mathbb{O}$.

5.1 Exceptional Jordan Algebra $J_3(\mathbb{O})$

Property	Value
$\dim(J_3(\mathbb{O}))$	$27 = 3^3$
$\dim(J_3(\mathbb{O})_0)$	26 (traceless)

E-series expression: 27 can be written in E-series dimensions:

$$\dim(J_3(\mathbb{O})) = \frac{\dim(E_8) - \dim(E_6) - \dim(SU_3)}{6} = \frac{248 - 78 - 8}{6} = \frac{162}{6} = 27$$

This is an arithmetic expression for 27 in E-series dimensions; $\dim J_3(\mathbb{O}) = 27$ is a Lie datum in its own right (main §3.1).

Lean: `j3o_e_series_certificate`.

5.2 F_4 Connection

F_4 is the automorphism group of $J_3(\mathbb{O})$:

$$\dim(F_4) = 52 = p_2^2 \times \alpha_{sum} = 4 \times 13$$

5.3 Exceptional Differences

Difference	Value	Framework expression
$\dim(E_8) - \dim(J_3(\mathbb{O}))$	$221 = 13 \times 17$	$\alpha_{\text{sum}} \times (\dim G_2 + N_{\text{gen}})$
$\dim(F_4) - \dim(J_3(\mathbb{O}))$	$25 = 5^2$	Weyl^2

Difference	Value	Framework expression
$\dim(E_6) - \dim(F_4)$	26	$\dim(J_3(\mathbb{O})_0)$

These are arithmetic rewritings (S2.7); they are not formalized as a group.

5.4 Structural Derivation of τ

The hierarchy parameter τ is defined as a ratio of framework constants:

$$\tau = \frac{\dim(E_8 \times E_8) \times b_2}{\dim(J_3(\mathbb{O})) \times H^*} = \frac{496 \times 21}{27 \times 99} = \frac{10416}{2673} = \frac{3472}{891}$$

Prime factorization:

- Numerator: $3472 = 2^4 \times 7 \times 31 = \dim(K_7) \times \dim(E_8 \times E_8)$
- Denominator: $891 = 3^4 \times 11 = {N_{gen}}^4 \times D_{bulk}$

Alternative form: $\tau_{num} = 7 \times 496 = \dim(K_7) \times \dim(E_8 \times E_8) = 3472$

τ is a chosen ratio of the constants. It is used only by Type II entries (S3), which are not counted.

Lean: `tau_structural_certificate`.

Part II: G_2 Holonomy Manifolds

6. Definition and Properties

6.1 G_2 as Exceptional Holonomy

Property	Value	Role
$\dim(G_2)$	14	Q_{Koide} numerator
$\text{rank}(G_2)$	2	Lie rank
Definition	$\text{Aut}(\mathbb{O})$	Octonion automorphisms

Lean: the G_2 cross-product statements are proved:

- `epsilon_antisymm`, `epsilon_diag`, `cross_apply` ✓
- `G2_cross_bilinear`, `G2_cross_antisymm`, `cross_self` ✓
- `G2_cross_norm` (Lagrange identity $\|u \times v\|^2 = \|u\|^2\|v\|^2 - \langle u,v\rangle^2$) ✓
- `reflect_preserves_lattice` (Weyl reflection) ✓
- `cross_is_octonion_structure` (case check over the 343 index triples) ✓
- `G2_equiv_characterizations` ✓ (by `rfl`: the two predicates are defined to coincide, so this is not a theorem about two characterizations)

6.2 G_2 as Kernel of the Lie Derivative

The G_2 subalgebra of so(7) admits a precise characterization as the stabilizer of the associative 3-form φ_0. For any antisymmetric matrix A in so(7), the Lie derivative of φ_0 is:

$$L_A(\varphi_0)_{ijk} = A_{ia}\varphi_{ajk} + A_{ja}\varphi_{iak} + A_{ka}\varphi_{ija}$$

The g_2 subalgebra consists of all A for which $L_A(\varphi_0) = 0$:

$$\mathfrak{g}_2 = \ker(L) = \{A \in \mathfrak{so}(7) : L_A(\varphi_0) = 0\}$$

This yields the decomposition so(7) = g_2 + V_7, where dim(g_2) = 14 and dim(V_7) = 7. The complement V_7 carries the standard 7-dimensional representation of G_2.

In practice, the kernel is computed via singular value decomposition (SVD) of the linear map L: so(7) –> $\Lambda^3(\mathbb{R}^7)$. The 14 singular vectors with eigenvalue zero span g_2; the 7 singular vectors with nonzero eigenvalue span V_7.

Note: A heuristic construction based on Fano-plane indices does not produce correct g_2 generators (each such generator is approximately 67% in g_2 and 33% in V_7). The kernel-based construction is the correct definition and must be used in all numerical computations involving g_2/V_7 decomposition.

6.3 Holonomy Classification (Berger)

Dimension	Holonomy	Geometry
7	**G_2**	**Exceptional**
8	Spin(7)	Exceptional

6.4 Torsion: Definition and Framework Interpretation

Mathematical definition: Torsion measures failure of G_2 structure to be parallel:

$$T = \nabla\phi \neq 0$$

For a G_2 structure φ, the intrinsic torsion decomposes into four irreducible G_2-modules:

$$T \in W_1 \oplus W_7 \oplus W_{14} \oplus W_{27}$$

Class	Dimension	Characterization
W_1	1	Scalar: $d\varphi = \tau_0 \star\varphi$
W_7	7	Vector: $d\varphi = 3\tau_1 \wedge \varphi$
W_{14}	14	Co-closed part of $d\star\varphi$

Class	Dimension	Characterization
W_{27}	27	Traceless symmetric

Total dimension: $1 + 7 + 14 + 27 = 49 = 7^2 = \dim(K_7)^2$

The torsion-free condition requires all four classes to vanish simultaneously, a highly constrained state with 49 conditions.

Torsion-free condition:

$$\nabla\phi = 0 \Leftrightarrow d\phi = 0 \text{ and } d * \phi = 0$$

What the framework uses. None of the 33 relations uses a torsion value. The integer ${\kappa_T}^{-1} = 61 = b_3 - \dim(G_2) - p_2$ enters three relations (α^{-1}, Y_p, Ω_m/Ω_Λ). No computation connects 61 to torsion: it is an integer combination without a physical name (register item R3).

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7. Topological Invariants

7.1 Combinations Used by the Dictionary

Constant	Formula	Value	Status
det(g)	$p_2 + 1/(b_2 + \dim(G_2) - N_{gen})$	65/32	imposed normalization (R2)
κ_T	$1/(b_3 - \dim(G_2) - p_2)$	1/61	integer combination (R3)
$\sin^2\theta_W$	$b_2/(b_3 + \dim(G_2))$	3/13	the $\sin^2\theta_W$ relation (S2.2)

det(g) and ${\kappa_T}^{-1}$ each have a second expression in the constants that agrees with the one above on the line $b_2 = 21$ and differs elsewhere (R2, R3; S2.1).

7.2 The 61 Decomposition

$$\kappa_T^{-1} = 61 = \dim(F_4) + N_{gen}^2 = 52 + 9$$

Alternative:

$$61 = p_2 \times \text{Weyl} \times (\text{Weyl} + 1) + 1 = 2 \times 5 \times 6 + 1$$

Further expressions for 61 are arithmetic facts of the kind recorded in S2.7; they do not give 61 a physical meaning (R3). Lean: `kappa_T_from_F4_generations`.

7.3 Continued Fraction Structure

The ratio $\dim(G_2)/H^*$ has the continued fraction

$$\frac{14}{99} = [0; 7, 14] = \cfrac{1}{7 + \cfrac{1}{14}}$$

whose partial quotients are $7 = \dim(K_7)$ and $14 = \dim(G_2)$. This is a property of the ratio 14/99: an arithmetic fact, with no physical consequence.

7.4 Pell Equation Structure

H^* and $\dim(G_2)$ satisfy a Pell equation:

$$\boxed{H^{*2} - 50 \times \dim(G_2)^2 = 1}$$

Explicitly:

$$99^2 - 50 \times 14^2 = 9801 - 9800 = 1$$

where $50 = \dim(K_7)^2 + 1 = 49 + 1$.

Fundamental solution: $(99, 14)$ is the fundamental solution of $x^2 - 50y^2 = 1$. It is the square of $\varepsilon = 7 + \sqrt{50}$, which solves $x^2 - 50y^2 = -1$:

$$\varepsilon^2 = (7 + \sqrt{50})^2 = 99 + 14\sqrt{50}$$

Continued fraction: $\sqrt{50} = [7; \overline{14}]$, with period 1, which is why the same two integers appear in §7.3.

Status: arithmetic identity, verified in Lean, with no physical consequence.

Part III: K_7 Manifold Construction

8. Twisted Connected Sum Framework

8.1 TCS Construction

The twisted connected sum (TCS) construction provides the primary method for constructing compact G_2 manifolds from asymptotically cylindrical building blocks.

Compact G_2 manifolds can be built by gluing two asymptotically cylindrical (ACyl) G_2 manifolds along their cylindrical ends, with the topology controlled by a twist diffeomorphism φ.

8.2 Asymptotically Cylindrical G_2 Manifolds

Definition: A complete Riemannian 7-manifold (M, g) with G_2 holonomy is asymptotically cylindrical (ACyl) if there exists a compact subset $K \subset M$ such that M K is diffeomorphic to $(T_0, \infty) \times N$ for some compact 6-manifold N.

8.3 Topological Status of (21, 77)

K_7 is posited with $(b_2, b_3) = (21, 77)$ (R1). **No construction of a compact G_2 manifold with these Betti numbers is currently known.** The Newton–Kantorovich certificate of [A] concerns a metric on the neck region; it does not certify the Betti numbers, and no global torsion-free G_2 structure has been constructed (main §9).

Catalogued constructions. The pair does not appear among the catalogued compact G_2 manifolds (Joyce's orbifold constructions; the twisted connected sums of Corti–Haskins–Nordström–Pacini). Orthogonal TCS is excluded by parity ($b_2 + b_3 = 98$ is even; CHNP Lemma 6.7). In a twisted connected sum, $b_2(M) = \mathrm{rk}\, K_+ + \mathrm{rk}\, K_- + \mathrm{rk}(N_+ \cap N_-)$, where $K\pm$ are the kernels of the restriction maps $H^2(Z\pm) \to H^2(K3)$ and $N\pm$ their images (CHNP); b_2 is therefore not bounded by the Picard rank of the K3 fibre, and $b_2 = 21 > 20$ does not by itself force a semi-Fano building block. Which building blocks and gluings could give $b_2 = 21$ is open. Non-orthogonal TCS and extra-twisted connected sums remain open within TCS.

No lattice decomposition 11 + 10. Sublattices N_1, N_2 of the K3 lattice $U^3 \oplus E_8(-1)^2$, of ranks 11 and 10, with $N_1 \oplus N_2 \oplus \langle 2 \rangle$ of full rank, do not support a TCS realisation. $N_1 \oplus N_2 \oplus \langle 2 \rangle$ cannot be an orthogonal decomposition of the unimodular K3 lattice ($\langle 2 \rangle$ is not unimodular), only a sublattice of finite index; and two building-block lattices $N_1 \perp N_2$ describe an orthogonal gluing, which is the case excluded by parity above. No embedding conditions are verified and no matching data is constructed. The split 11 + 10 is therefore not available.

A Joyce–Karigiannis route does not realise (21, 77) either (§8.4).

Global properties assumed, not established:

- Compact 7-manifold (no boundary)
- Simply connected: $\pi_1(K_7) = \{1\}$, $b_1 = 0$
- If a torsion-free G_2 structure g* exists on K_7, then $\mathrm{Hol}(g^*) = G_2$ [20, Prop. 11.2.3] and $\mathrm{Ric}(g^*) = 0$. Its existence is open (main §9; [ERR, §7.5]).
- (Euler characteristic $\chi(K_7) = 0$ is automatic for a closed odd-dimensional manifold; §9.1.)

Arithmetic. $b_2 = 21 = C(7,2)$ and $b_3 = 77 = C(7,3) + 2b_2 = 35 + 42$ are arithmetic facts about the posited integers; they define no geometric decomposition (§9.3).

8.4 The Joyce–Karigiannis $\mathbb{Z}_2{}^3$ Route Does Not Realise (21, 77)

The Joyce–Karigiannis framework [43] resolves orbifolds $(T^3 \times K3)/G$, with $G \subset \mathrm{Aut}(T^3) \times \mathrm{Aut}(K3)$ acting with A_1-type singularities, by gluing families of Eguchi–Hanson spaces, and yields compact torsion-free G_2 manifolds. For a resolution of this type the Betti numbers are

$$b_2(N) = b_2\big((T^3 \times K3)/G\big) + b_0(\text{fixed loci}), \qquad b_3(N) = b_3\big((T^3 \times K3)/G\big) + b_1(\text{fixed loci}).$$

The route. Take $G = \mathbb{Z}_2{}^3$ with a mixed-parity action, on a K3 whose lattice action exists abstractly by Mukai [44], Garbagnati–Sarti [45] and Nikulin [51], with a fixed-locus table contributing $(b_0, b_1) = (21, 55)$. The route reaches $b_2 = 0 + 21 = 21$ and $b_3 = 22 + 55 = 77$ only if the quotient cohomology is $(b_2, b_3) = (0, 22)$, on a K3 of Picard rank 1.

Why it fails. The quotient cohomology is incompatible with the action (audit of 2026-09-16 in the repository Arithmon/K7, `docs/openwave-candidate/K7_P2_JK_postfreeze_cohomology_audit_2026_09_16.md`, with the checker `check_k3_jk_quotient_cohomology.py`):

- G acts diagonally on $T^3 \times K3$, so the rational cohomology of the quotient is the G-invariant part of the cohomology of $T^3 \times K3$. The trace of G on $H^2(K3)$ gives character multiplicities (5, 5, 2, 2, 2, 2, 2, 2), so $\dim H^2(K3; \mathbb{Q})^G = 5$.
- With $W = H^1(T^3; \mathbb{Q})$ a sum of three characters and $H^1(K3) = H^3(K3) = 0$, the Künneth formula gives $b_2((T^3 \times K3)/G) = 5 + \#\{i < j : \chi_i = \chi_j\} \geq 5$ and $b_3((T^3 \times K3)/G) \leq 16$. The checker enumerates all $8^3 = 512$ ordered torus-character triples and confirms both bounds; the product action compatible with the G_2 form gives $(b_1, b_2, b_3) = (1, 6, 16)$.
- The pair (0, 22) is therefore impossible for this action. Keeping the fixed-locus contributions (21, 55) and the same resolution formula would give $b_2 \geq 26$ and $b_3 \leq 71$, not (21, 77).
- The involution profile also requires a K3 of Picard rank at least 17 (`check_k3_jk_character_gate.py`; `K7_P2_neutrino_JK_character_obstruction.md`), not Picard rank 1.

Scope. This closes the route as stated. It does not show that every Joyce–Karigiannis construction, or every construction of a compact G_2 manifold with Betti numbers (21, 77), is impossible. Repairing the route would require a fully specified product action, its quotient Betti numbers computed directly, and an independent audit of the singular components and of the resolution formula.

The neck. The Newton–Kantorovich-certified neck metric of [A] and the closed-form neck ansatz of [D] operate at the neck level of a putative compact extension and do not depend on which global construction, if any, completes it. They are evidence about a neck, not about the existence of K_7.

Intermediate-curvature splitting (Chen–Hong 2026). Chen and Hong [48] prove that, for certain (n, m), a noncompact manifold with nonnegative m-intermediate curvature admitting a smooth proper map of nonzero degree to $M^{n-m} \times T^{m-1} \times \mathbb{R}$ splits isometrically; outside that range they construct non-split examples on $S^{n-m} \times T^{m-1} \times \mathbb{R}$. For the neck topology $K3 \times T^2 \times \mathbb{R}$ ($n = 7$, $m = 3$) the splitting theorem does not apply. Nothing follows about the neck: being outside the range of a rigidity theorem does not show that non-split metrics with this curvature condition exist on $K3 \times T^2 \times \mathbb{R}$ (the examples are on $S^4 \times T^2 \times \mathbb{R}$). In particular it does not follow that the Chebyshev corrections of the neck metric [A] are free of a forced splitting.

9. Cohomological Structure (Conditional on Existence)

9.1 Betti Spectrum, Poincaré Duality and Euler Characteristic

If a compact, simply connected K_7 with holonomy G_2 and $(b_2, b_3) = (21, 77)$ exists, Poincaré duality fixes its full Betti spectrum:

k	0	1	2	3	4	5	6	7
$b_k(K_7)$	1	0	21	77	77	21	0	1

with $b_1 = 0$ from simple connectivity, b_2 and b_3 posited (R1), and b_4 to b_7 by duality. The Euler characteristic of any compact oriented odd-dimensional manifold vanishes: $\chi(K_7) = 0$. Lean: `euler_char_K7_is_zero`, `poincare_duality_K7`. The combination $H^* = b_2 + b_3 + 1 = 99$ is used by the dictionary.

A Mayer–Vietoris computation of (b_2, b_3) would require a decomposition of K_7, for example the two halves of a twisted connected sum; none is available (§8.3).

9.2 Numerical Consistency

The spectral analysis of the approximate neck metric ([B]) reports 21 near-zero eigenvalues of the Laplacian on 2-forms and 77 on 3-forms, separated from the rest of the spectrum. These multiplicities are not independent evidence for the posited Betti numbers: [B] constructs the 2-forms from the split 11 + 10, which is not available (§8.3), and the 3-forms from the split 35 + 21 + 21, whose block of 35 is not a subspace of the cohomology (§9.3). Nor are they a certificate, since the metric is approximate and is not a metric on a compact K_7.

9.3 The Integer 42

The number 42 appears in several relations as a combination of framework constants:

Expression	Value
$p_2 \times N_{gen} \times \dim(K_7)$	$2 \times 3 \times 7 = 42$
$2 \times b_2$	$2 \times 21 = 42$
$b_3 - C(7,3)$	$77 - 35 = 42$

It is not the Euler characteristic ($\chi(K_7) = 0$). Lean: `magic_42_gift_form`; at commit `34ce857` 42 is `two_b2 := 2 * b2`, and the earlier name `chi_K7` is kept only as a deprecated alias (S3.7).

No "local ⊕ global" splitting of H^3. The identity 35 + 42 = 77 does not give a splitting $H^3(K_7) = H^3_{local} \oplus H^3_{global}$ with the 35 identified with $\Lambda^3(\mathbb{R}^7)$. The 35 is the rank of the bundle $\Lambda^3 T^*K_7$, a pointwise dimension; it does not provide a 35-dimensional subspace of the global cohomology $H^3(K_7)$. The identity is arithmetic.

Part IV: Metric Structure and Verification

10. Structural Metric Invariants

10.1 Structural Metric Invariants and Normalizations

Two quantities that the dictionary uses are written as properties of a metric. Neither is computed from a metric: $\kappa_T{}^{-1}$ is an integer combination and det(g) is a normalization imposed on the metric.

Invariant	Formula	Value	Status
κ_T	$1/(b_3 - \dim(G_2) - p_2)$	1/61	integer combination (R3)
det(g)	$p_2 + 1/(b_2 + \dim(G_2) - N_{gen})$	65/32	imposed normalization (R2; §10.3)

10.2 The Combination $\kappa_T = 1/61$

Arithmetic:

$$\kappa_T = \frac{1}{b_3 - \dim(G_2) - p_2} = \frac{1}{77 - 14 - 2} = \frac{1}{61}$$

Nothing in the framework identifies b_3 with fermion modes or $\dim(G_2)$ with gauge constraints in this subtraction.

Status: integer combination (R3)

10.3 Metric Determinant det(g) = 65/32

The value 65/32 can be written in several ways in the framework constants; three are:

Expression 1:

$$\det(g) = \frac{\text{Weyl} \times (\text{rank}(E_8) + \text{Weyl})}{2^{\text{Weyl}}} = \frac{5 \times 13}{32} = \frac{65}{32}$$

Expression 2 (the one used, main App. B):

$$\det(g) = p_2 + \frac{1}{b_2 + \dim(G_2) - N_{\text{gen}}} = 2 + \frac{1}{21 + 14 - 3} = 2 + \frac{1}{32} = \frac{65}{32}$$

Expression 3:

$$\det(g) = \frac{H^* - b_2 - 13}{32} = \frac{99 - 21 - 13}{32} = \frac{65}{32}$$

Expressions 1 and 2 agree on the line $b_2 = 21$ and differ elsewhere (R2); expression 3 uses only b_3. The order of events is this: the metric optimization was constrained to det(g) = 65/32 (a normalization target

chosen to be compatible with the metric components $g_{ss} = 19/6$ and ${g_T}^2 = 7/6$ of [A]), and the formulas above were identified afterwards. Any rational number with a small numerator and denominator can be written as combinations of a few integers.

Numerical value: $65/32 = 2.03125$ (exact rational)

Status: imposed normalization (R2), exact by construction. Six of the 33 relations use $\det_{num}$ or $\det_{den}$ and depend on this choice (main §4.4).

---

11. Reference Form and Formalization

11.1 The Algebraic Reference Form

Scaling law. A positive 3-form φ determines its metric g_φ through

$$(\iota_x\varphi) \wedge (\iota_y\varphi) \wedge \varphi = 6\, g_\varphi(x, y)\, \mathrm{vol}_{g_\varphi}.$$

Under $\varphi \mapsto c\varphi$ ($c > 0$) the left side scales by c^3, and if $g \mapsto \lambda g$ the volume form scales by $\lambda^{7/2}$; hence $c^3 = \lambda^{9/2}$, that is

$$g_{c\varphi} = c^{2/3}\, g_\varphi, \qquad \det g_{c\varphi} = c^{14/3}\, \det g_\varphi.$$

The reference form. Fix a coframe $(e^1, \ldots, e^7)$ and let φ_0 be the standard form in it (§12.1); in that coframe the matrix of g_{φ_0} is I_7. To impose $\det g = D = 65/32$ in this fixed coframe,

$$\boxed{\varphi_{\mathrm{ref}} = D^{3/14}\, \varphi_0, \qquad g_{\mathrm{ref}} = D^{1/7}\, g_{\varphi_0}, \qquad \det g_{\mathrm{ref}} = D.}$$

Check. The exponent is $3/14$, not $1/14$: by the scaling law, $D^{1/14}\ \varphi_0$ would induce $\det g = D^{1/3} \approx 1.26645$, not $65/32$.

Which metric the coframe is orthonormal for. The coframe is orthonormal for g_{φ_0}. For g_{ref} it is not: the matrix of g_{ref} in it is $D^{1/7}\ I_7$. In a coframe orthonormal for g_{ref} itself, the matrix of g_{ref} is I_7 and the form reads φ_0 again; the normalization $\det g = 65/32$ is a statement relative to the fixed coframe, not an invariant of the G_2 structure. "I_7" reflects this frame choice, not global flatness. The Newton–Kantorovich-certified model of [A] is a conditional statement about a neck region; it does not construct a compact K_7 (main §9).

Property	Value	Status
det(g)	65/32	exact by imposed normalization (§10.3)
φ_{ref} components	7 of 35 non-zero	standard form (§12.2)
Joyce smallness hypothesis	depends on geometric bounds not established here	not verified (§11.2)

11.2 Joyce Existence Theorem and Global Solutions

Clarification: The reference form $\varphi_{\text{ref}} = \mathrm{D}^{3/14}\,\varphi_0$ is the standard form written in the fixed coframe of §11.1 (orthonormal for g_{φ_0}, not for g_{ref}), not a globally constant form on K_7. On a compact G_2 manifold, the coframe 1-forms $\{e^i\}$ satisfy $de^i \neq 0$ in general, so "constant components" does not imply $d\varphi = 0$ globally.

Actual solution structure: The topology and geometry of K_7 impose a deformation:

$$\varphi = \varphi_{\text{ref}} + \delta\varphi$$

The torsion-free condition ($d\varphi = 0$, $d^*\varphi = 0$) is a **global constraint**. Joyce's perturbation theorem gives a nearby torsion-free G_2 metric when the torsion of a closed G_2 structure is small compared with a constant that depends on the injectivity radius and curvature bounds of that structure ([20], Thm. 11.6.1). There is no universal threshold: the constant has not been computed for this structure, so a small numerical torsion value is not an existence argument (main §9). Nor does the number $\kappa_T = 1/61$ bear on Joyce's smallness hypothesis [ERR, §7.6].

11.3 Lean 4 Formalization

Scope of verification: The Lean formalization (Arithmon/K7-Lean; fourteen axioms, no `sorry`) verifies:

1. Arithmetic identities and algebraic relations between the framework's constants, including the 33 relations (`relations_v36_certified`, standard axioms only; S3.7)
2. Numerical enclosures (the ten interval certificates)
3. G_2 linear algebra on $\mathbb{R}^7$: exterior algebra, Hodge star, $\psi = \star\varphi$ (`Geometry` module)
4. Spectral modules (arithmetic of a first-eigenvalue formula), which no relation uses

The axioms (see main §8): four axioms import literature results (`cheeger_inequality`, `literature_package`, `spectral_upper_bound`, `neck_dominates`), and ten interval certificates import numerical enclosures (`det_g_at_half`, `K3_eigenvalue_0...3` and their `_bracketed` forms). They carry content that cannot be checked by `decide` alone because it involves real arithmetic [ERR, §7.1].

Outside its scope: the existence of K_7 as a smooth G_2 manifold, and any physical interpretation of the invariants.

```
-- GIFT.Foundations.AnalyticalMetric

def phi0_indices : List (Fin 7 × Fin 7 × Fin 7) :=
  [(0,1,2), (0,3,4), (0,5,6), (1,3,5), (1,4,6), (2,3,6), (2,4,5)]

def phi0_signs : List Int := [1, 1, 1, 1, -1, -1, -1]

def scale_factor_power_14 : Rat := 65 / 32
```

```
theorem torsion_satisfies_joyce :
  torsion_norm_constant_form < joyce_threshold_num := by native_decide

theorem det_g_equals_target :
  scale_factor_power_14 = det_g_target := rfl
```

Both statements are true as written and say less than their names. `torsion_satisfies_joyce` compares two integers, one of them a fixed threshold (288) that is not Joyce's constant, which has not been computed for this structure (§11.2); it is not an existence argument. `det_g_equals_target` is a definitional equality of the rational 65/32 with the target (§12.5).

11.4 The Chain of Choices

The logical structure from algebra to the reference metric, and where the framework's choices enter:

```
Octonions (O)
     |
     ▼
G_2 = Aut(O), dim = 14
     |
     ▼
Standard form φ_0 (Harvey-Lawson 1982)
     |
     ▼
φ_ref = D^{3/14} φ_0, D = 65/32  ← imposed normalization (R2)
     |
     ▼
Metric g_ref = D^{1/7} g_{φ_0}    (law g_{cφ} = c^{2/3} g_φ)
     |
     ▼
det(g) = 65/32               ← imposed (§10.3)
     |
     ▼
sin^2θ_W = 3/13, Q = 2/3, ...  ← assigned (main §3); the metric enters none of them
```

---

12. Analytical G_2 Metric Details

12.1 The Standard Form φ_0

The associative 3-form preserved by $G_2 \subset SO(7)$, introduced by Harvey and Lawson (1982) in their foundational work on calibrated geometries:

$$\varphi_0 = \sum_{(i,j,k)\in\mathcal{I}} \sigma_{ijk}\, e^{ijk}$$

where:

- $\mathcal{I}$ = {(0,1,2), (0,3,4), (0,5,6), (1,3,5), (1,4,6), (2,3,6), (2,4,5)}
- σ = (+1, +1, +1, +1, -1, -1, -1)

12.2 Linear Index Representation

In the C(7,3) = 35 basis:

Index	Triple	Sign	Index	Triple	Sign
0	(0,1,2)	+1	23	(1,4,6)	-1
9	(0,3,4)	+1	27	(2,3,6)	-1
14	(0,5,6)	+1	28	(2,4,5)	-1
20	(1,3,5)	+1	–	–	–

All other 28 components are exactly 0.

12.3 Metric Derivation

The metric of a positive 3-form is defined intrinsically by the identity of §11.1. Equivalently, with

$$B_\varphi(x,y) = \tfrac{1}{6}\,(\iota_x\varphi)\wedge(\iota_y\varphi)\wedge\varphi \in \Lambda^7,$$

written in the fixed coframe as $B_{ij}\ e^{1\cdots7}$, one has $g_\varphi = (\det B)^{-1/9}\ B$. For the standard φ_0, $B = I_7$ and $g = I_7$, $\det g = 1$; for $c\varphi_0$, $B = c^3\ I_7$ and $g = c^{2/3}\ I_7$, recovering the scaling law.

The contraction $g_{ij} = (1/6)\ \Sigma_{k,l}\ \varphi_{ikl}\ \varphi_{jkl}$ raises indices with a background Euclidean metric, so it is not a general reconstruction of g_φ: it agrees with g_φ only when φ is already standard for that background metric, and for $c\varphi_0$ it gives $c^2\ I_7$ instead of $c^{2/3}\ I_7$. Setting $c^{14/3} = 65/32$, i.e. $c = (65/32)^{3/14}$, yields the framework's reference metric.

12.4 Comparison: Fano Plane vs G_2 Form

Structure	7 Triples	Role
Fano lines	(0,1,3), (1,2,4), (2,3,5), (3,4,6), (4,5,0), (5,6,1), (6,0,2)	G_2 cross-product ε_{ijk}
G_2 form	(0,1,2), (0,3,4), (0,5,6), (1,3,5), (1,4,6), (2,3,6), (2,4,5)	Associative 3-form

Both have 7 terms because both are Fano-plane patterns: the associative calibration is $\varphi(x, y, z) = \langle x \times y, z\rangle$ on $\mathrm{Im}(\mathbb{O})$, and the two tables differ by a relabelling of the seven points.

12.5 Verification Summary

Method	Result	Reference
Algebraic	$\varphi_{\mathrm{ref}} = (65/32)^{3/14}\ \varphi_0$, $g_{\mathrm{ref}} = (65/32)^{1/7}\ g_{\varphi_0}$	§11.1
Lean 4	`det_g_equals_target :` `rfl` (a definitional equality of the rational 65/32 with the target; it does not check the scaling law)	AnalyticalMetric.lean
Joyce theorem	applies under a smallness hypothesis whose constant is not computed here	[20]; main §9

None of these methods establishes a torsion-free metric on a compact K_7 (main §9).

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13. Changes since Version 3.5

The sections above state the present position. This section lists, once, what versions up to 3.5 of this supplement said that version 3.6 no longer says. The corrections of record are in the erratum [ERR].

Earlier statement	Present position	Section
the "Weyl Triple Identity", evidence that Weyl = 5 is "structurally constrained"; a "cosmological ratio $\sqrt{}$Weyl" (from the relation $\Omega_m = \Omega_{DE}/\sqrt{5}$)	an arithmetic fact; the relation is not among the 33 (S2.6)	§2.3
the "Exceptional Chain Theorem"	three factorizations with an index chosen case by case	§3.2
a "triple geometric origin" of p_2; p_2 as a Pontryagin class	occurrences of the number 2; the binary ratio $\dim(G_2)/\dim(K_7)$ [ERR, §6]	§4.3
τ established as "a geometric constant rather than a free parameter"	a chosen ratio	§5.4

Earlier statement	Present position	Section
$\kappa_T = 1/61$ as a "topological capacity for torsion"; 61 as "effective matter degrees of freedom"; physical interactions "emerge from the geometry of K_7"; $\kappa_T = 1/61$ places K_7 within Joyce's perturbative regime	an integer combination without a physical name [ERR, §7.6]	§6.4, §10.2, §11.2
the near-identity $\dim(G_2)/\sqrt{2} \approx \pi^2$ (0.30%) "resolved" by the closed form for λ_1; a Lean module `Spectral.Selberg Bridge` connecting the spectral gap to a Dirichlet polynomial through the Selberg trace formula; the Pell identity combined with a "selection principle" $\kappa = \pi^2/14$ (Lean module `Spectral.Selection Principle`) into a link with the spectral gap	none of these bears on the framework; the section on spectral geometry is removed	§7.4
a building-block scan showing that $b_2 = 21 > 20 = \max \rho(K3)$ forces a semi-Fano building block	b_2 is not bounded by the Picard rank of the fibre	§8.3
sublattices of ranks 11 and 10 "consistent with a TCS construction"	the split 11 + 10 is not available	§8.3
a Joyce–Karigiannis $\mathbb{Z}_2{}^3$ orbifold [43] realises (21, 77), by a four-phase computer-assisted audit (a symplectic V4 screen on a complete-intersection K3, an anti-symplectic obstruction for P^5-linear maps, the abstract lattice action, a fixed-locus table) and a quotient cohomology (0, 22) on a K3 of Picard rank 1; the Lean module `Joyce KarigiannisConstruction.lean` (K7-Lean commit `6acbf49`, since removed; theorem `jk_z23_ construction_realizes_gift_betti`) as evidence	the route does not realise the pair; the module verified the arithmetic of the Betti formula on an erroneous input	§8.4
by Chen–Hong [48], the corrections of the neck metric are "not obstructed by a forced splitting"	nothing follows about the neck	§8.4
$H^3(K_7) = H^3{}_{\text{local}} \oplus H^3{}_{\text{global}}$ with dimensions 35 + 42	no such splitting	§9.3

Earlier statement	Present position	Section
$\varphi_{\text{ref}} = c\ \varphi_0$ with $c = D^{1/14}$ and $g_{\text{ref}} = c^2\ I_7$, from the law $g_{c\varphi} = c^2\ g_\varphi$ and the contraction formula for the metric	$\varphi_{\text{ref}} = D^{3/14}\ \varphi_0$, from the law $g_{c\varphi} = c^{2/3}\ g_\varphi$; the metric $D^{1/7}\ I_7$ is unchanged	§11.1, §12.3
a neural-network torsion value compared with a fixed threshold $\varepsilon_0 = 0.1$, a "224× safety margin"	no universal threshold; Joyce's constant is not computed	§11.2
a section of numerical experiments with neural networks: torsion values, a `g2_score` described as a measure of holonomy, and det(g) = 65/32 "passing 8/8 independent consistency tests"	the section is removed: the scores were training diagnostics, not a measure of holonomy, and det(g) is an imposed normalization (R2)	§10.3
four "external data-package" axioms	fourteen axioms: four literature imports and ten interval certificates [ERR, §7.1]	§11.3

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Author's Related Works

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The K_7 Framework (formerly GIFT): Supplement S1 Mathematical Foundations: E_8 + G_2 + K_7

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